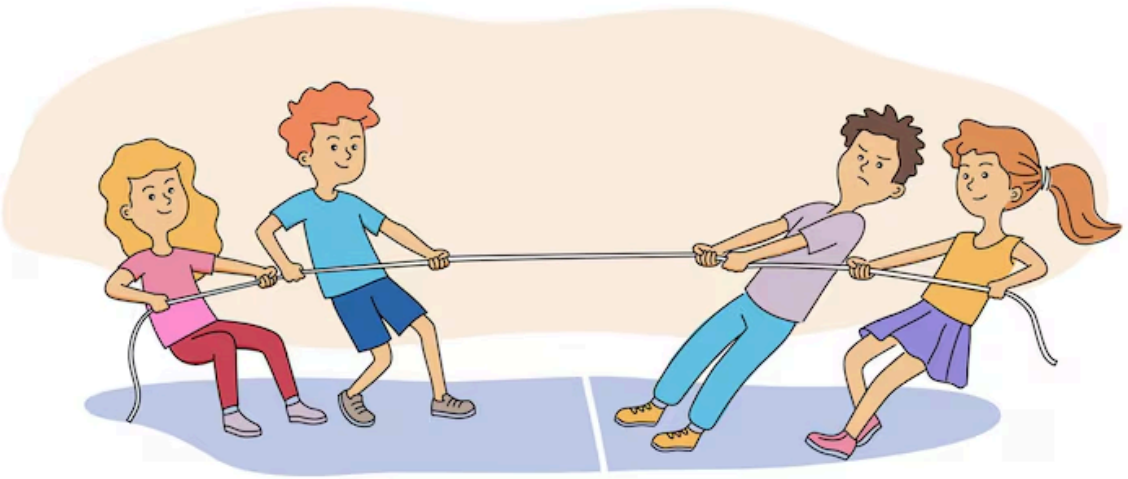
Full Stack Intern Assignment: Interactive Number Game
Deadline for the submission of the assignment : **before 05/05/2026 11:00 pm**
Mail the submissions : vinay@vidh.co , cc: ram@vidh.co

Goal

Design and develop a two-player interactive number game optimized for a horizontal tablet/phone view (mocked as a "teaching panel"). The game features a "Tug-of-War" style mechanic where players answer equations, and the winner of each round pulls the rope animation toward their side based on points. The overall winner is declared after a set number of questions, followed by a confetti celebration.



Target Audience

Two players (e.g., students, peers) using a touch-enabled device (tablet or large horizontal phone screen).

Core Features and Requirements

1. User Interface (UI) and Layout

- **Horizontal Optimization:** The entire layout must be designed for landscape/horizontal orientation, maximizing screen real estate for the game board.

- **Two-Player Split:** The screen should be clearly divided, assigning one half to Player 1 and the other half to Player 2.
- **Teaching Panel Mockup:** The design should be clean, large, and touch-friendly, simulating an application for an educational/teaching device.
- **Responsive Design:** The UI should scale appropriately for different horizontal screen sizes.

2. Game Mechanics

A. Question Generation

- **Equation Type:** Simple arithmetic equations (addition, subtraction, multiplication, and optionally division).
- **Difficulty:** Questions should be generated randomly with adjustable difficulty (e.g., single-digit, double-digit operations).
- **Display:** The equation should be centrally displayed, easily visible to both players.

B. Input and Interaction

- **Simultaneous Input:** Both players should have dedicated input fields (or number pads) on their respective sides to enter their answers quickly.
- **Timing:** Implement a simple timer per question (e.g., 10-20 seconds) to encourage quick responses.
- **Lock-in:** Once a player submits an answer, their input should be locked for that round.

C. Scoring and Rope Animation

- **Round Winner:** The first player to submit the **correct** answer wins the round.
- **Point Allocation:**
    - Round Winner: 1 point.
    - Incorrect/Timeout: 0 points.
- **Rope Animation:** A central "rope" graphic must visually represent the score difference (Tug-of-War).
    - When Player 1 wins, the rope animation moves toward Player 1's side.
    - When Player 2 wins, the rope animation moves toward Player 2's side.
    - The movement distance should be proportional to the points won.

3. Game Flow and Ending

- **Total Questions:** The game should run for a predetermined number of questions (e.g., 5 or 10 rounds).
- **Winner Determination:** The player with the highest total score after the final question is declared the winner.
- **Confetti Animation:** Upon declaring the winner, a celebratory confetti animation should fill the screen.

Deliverables :

Task	Description
Project Setup	Initialize project structure (HTML, CSS, JS files).
Horizontal Layout	Implement a two-column, landscape-optimized layout using CSS Flexbox/Grid.
Player Areas	Create distinct areas for Player 1 and Player 2 (Score display, Input field/Numpad).

Task	Description
Equation Generator	Function to generate random arithmetic problems.
Input Handling	Event listeners for answer submission from both players.
Round Logic	Logic to check correctness, determine the winner (first correct answer), and update the score.
Rope Animation Logic	State management to control the rope's position based on the score difference.

Task	Description
Game Flow Control	Implement the main loop to manage the set number of questions.
Winner Screen	Display the final winner's name prominently.
Confetti Effect	Implement a visual confetti animation upon game completion.